EXPERIMENT 5 : CRYPT ARITHMETIC PROBLEM

AIM

To develop a Python program that solves the Crypt Arithmetic Problem using a brute-force search by assigning unique digits to letters and finding a valid solution that satisfies the given arithmetic equation (SEND + MORE = MONEY).

Objective, algorithm,result

OBJECTIVE

- To understand the concept of the Crypt Arithmetic Problem in Artificial Intelligence.
- To assign unique digits to letters without repetition.
- To solve the equation SEND + MORE = MONEY.
- To verify that the assigned digits satisfy the arithmetic expression.

ALGORITHM

Step 1: Start.

Step 2: Define the letters in the puzzle (S, E, N, D, M, O, R, Y).

Step 3: Generate all possible unique digit assignments (0–9) for these letters.

Step 4: Ensure that the leading letters (S and M) are not assigned zero.

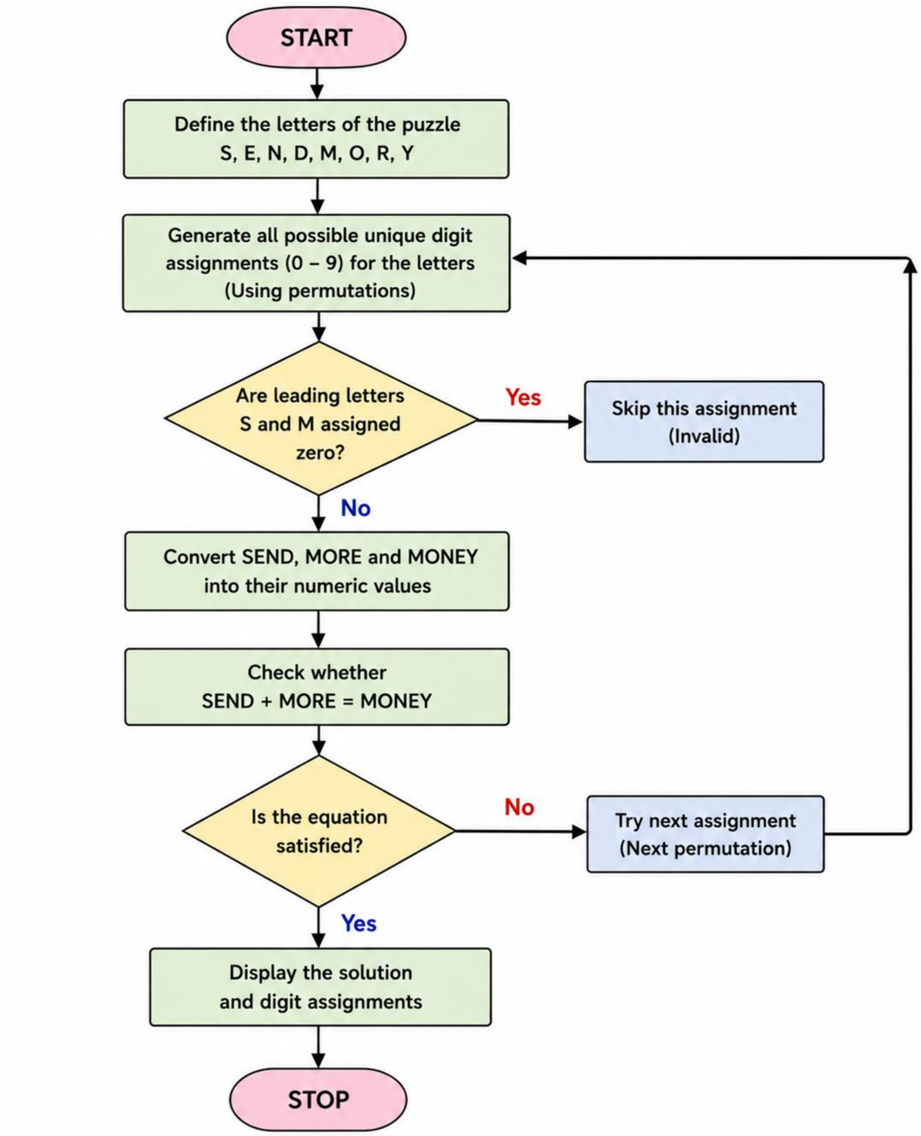
Step 5: Convert the words SEND, MORE, and MONEY into their corresponding numeric values.

Step 6: Check whether the equation SEND + MORE = MONEY is satisfied.

Step 7: If the equation is true, display the solution and the digit assignments.

Step 8: Stop.

FLOW CHART



SOURCE CODE

```
from itertools import permutations


letters = ('S', 'E', 'N', 'D', 'M', 'O', 'R', 'Y')
```

```
digits = range(10)

for perm in permutations(digits, len(letters)):
    S, E, N, D, M, O, R, Y = perm

    # Leading digits cannot be zero
    if S == 0 or M == 0:
        continue

    SEND = 1000*S + 100*E + 10*N + D
    MORE = 1000*M + 100*O + 10*R + E
    MONEY = 10000*M + 1000*O + 100*N + 10*E + Y

    if SEND + MORE == MONEY:
        print("Solution Found:")
        print("SEND =", SEND)
        print("MORE =", MORE)
        print("MONEY =", MONEY)
        print("\nLetter Assignments:")
        print("S =", S)
        print("E =", E)
        print("N =", N)
        print("D =", D)
        print("M =", M)
        print("O =", O)
        print("R =", R)
        print("Y =", Y)
```

break

OUTPUT

```
y
Solution Found:
SEND = 9567
MORE = 1085
MONEY = 10652

Letter Assignments:
S = 9
E = 5
N = 6
D = 7
M = 1
O = 0
R = 8
Y = 2
```

RESULT

The Python program successfully solves the Crypt Arithmetic Problem by assigning unique digits to each letter such that the equation SEND + MORE = MONEY is satisfied. The valid solution obtained is:

- SEND = 9567
- MORE = 1085
- MONEY = 10652

Thus, the crypt arithmetic puzzle is solved successfully.